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Paleomagnetic evidence for tectonic rotation of northern Mexico and the continuity of the Cordilleran orogenic belt between Nevada and Chihuahua

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ABSTRACT

Results of paleomagnetic studies of volcanic and sedimentary rocks from Chihuahua, northern Mexico, suggest that a counterclockwise tectonic rotation relative to "stable" North America occurred during early Tertiary time. The results permit development of a model for the tectonic evolution of southern North America, in which northern Mexico south of the Texas lineament is rotated in order to restore it to its Eocene-Oligocene position. This explains the "anomalous" zone of the Cordilleran orogenic belt between southern Nevada and northern Chihuahua, which suggests that the belt was once continuous and that the complications are the result of later tectonic events. The restoration may also explain the dislocation of the Mesozoic batholith belt of western Mexico, and the large nappe structures of the Sierra Madre Oriental within the continuation of the belt into Mexico.

INTRODUCTION

All hypotheses concerned with the tectonic evolution of southern North America are hindered by the lack of data from Mexico. Thus, studies concerned with large-scale features of the southern United States typically do not address the question of continuation into Mexico. One of the major geologic features of North America is the Cordilleran orogenic belt (King, 1969, 1977; Drewes, 1978). The belt forms a remarkable feature of the Pacific margins of North and South America. The belt has long been regionally and locally studied, and it is regarded as the result of a major compressive orogeny during Cretaceous–early Tertiary time (Drewes, 1978; King, 1969; Coney, 1978). The belt clearly extends into Mexico, but between southern Nevada and northern Chihuahua (Fig. 1), its continuity is not so readily apparent as elsewhere. Albritton and Smith (1957) suggested that the belt may have been displaced left laterally by some 1,000 km along the Texas lineament. Drewes (1978) suggested that the orogenic belt is continuous, without major interruptions, but locally deformed between a prism of miogeosynclinal rocks and basement crystalline rocks with smaller tectonic displacements of about 100 km (Drewes, 1978). This paper reports new paleomagnetic data of Eocene rocks from an area in central Chihuahua, which lies southwest of the Texas lineament and the problematic zone of the Cordilleran orogenic belt and is within the continuation of the belt into Mexico (Fig. 1).

PALEOMAGNETIC DATA

The rocks sampled come from the San Antonio Camp of the eastern part of the Santo Domingo–San Antonio mining district, which has been worked since the early 1700s (Syner and Hewitt, 1952). The basement rocks at a depth of some 1,750 m are represented by a quartz monzonite, which is overlain by an evaporite series (some 180 m thick), then the Cretaceous limestone (some 1,260 m thick, divided into four informal horizons), which is intruded by several igneous bodies, and finally the capping sequence (about 290 m thick), which is separated from the Cretaceous limestone by a rugged erosion surface (Spurr, 1911; Hewitt, 1943, 1966). The rock samples described are from the capping sequence, which includes interbedded rhyolite flows, tuffs, and conglomerates (Fig. 2). From paleontological evidence presented by Hewitt (1966), the sequence is considered to be of early Tertiary age, probably Eocene. The paleomagnetic samples were collected from the eastern part of the mining district, the San Antonio Camp, and come from the rhyolite flows and tuffs; one site (SA 4) is in the matrix of the first conglomerate (Fig. 2). The sampling scheme was designed to cover most of the time span available, and a total of 12 sites were drilled in the field. The natural remanent magnetization (NRM) of 65 specimens (2.54 cm diameter and height) cut from 36 cores was measured on a DIGICO spinner magnetometer. For each site, two to four specimens representing typical NRM

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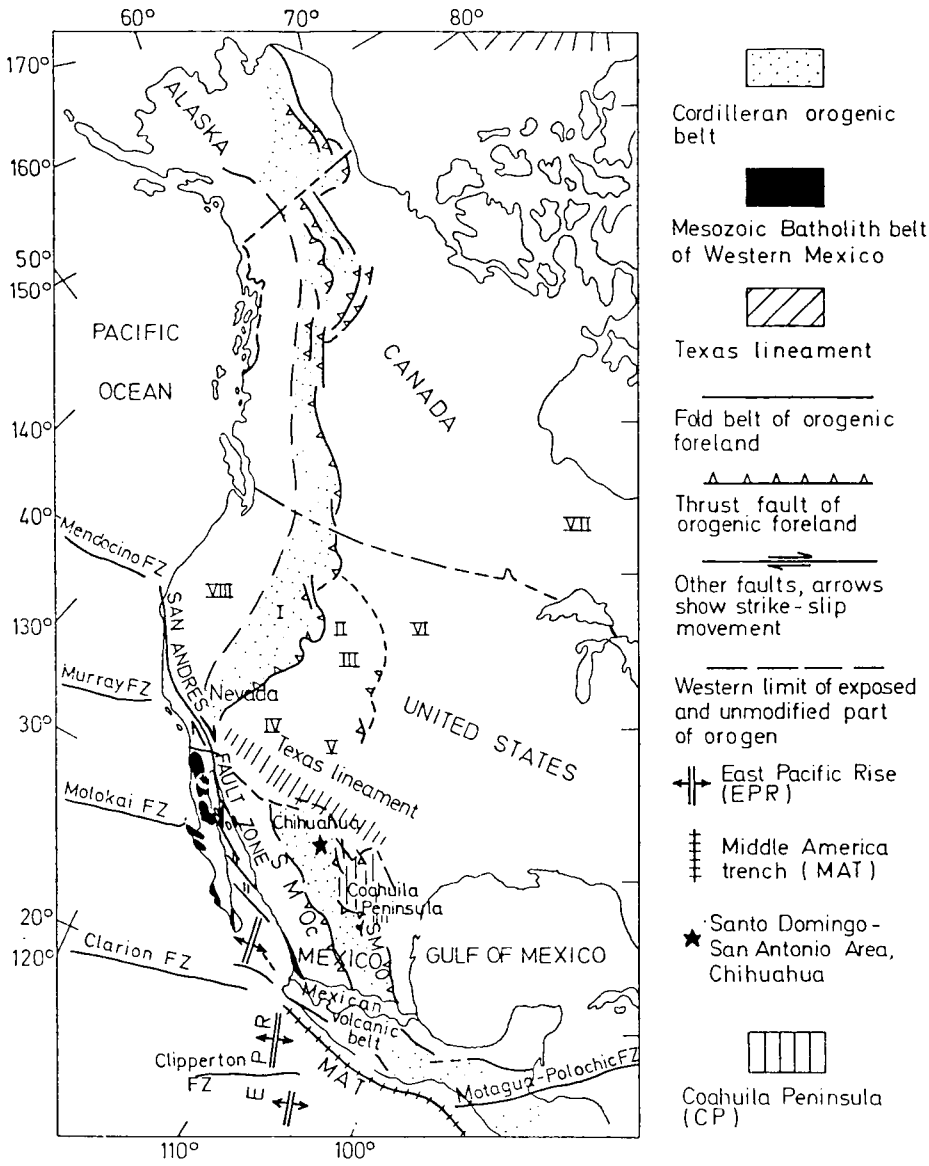


Figure 1. Schematic map of North America showing location of (I) Cordilleran orogenic belt; (II) Rocky Mountains; (III) Colorado Plateau; (IV) Basin and Range province; (V) Rio Grande rift; (VI) Interior Lowlands; (VII) craton; (VIII) zone of younger deformation, metamorphism, volcanism, and sedimentation. Star indicates location of area sampled for paleomagnetic studies. MAT, Middle America Trench; SMO, Sierra Madre Oriental; SMOc, Sierra Madre Occidental; FZ, fracture zone. Modified after Drewes (1978), King (1969), and Tardy (1975).

properties (intensity and direction) were used as pilot specimens for incremental partial thermal cleaning (up to 605 °C) and alternating field (AF) demagnetization (up to 950 Oe). All specimens carried stable high-coercivity components of remanence, generally with a weak low coercivity component, most probably of a viscous nature. Stability index analyses (Giddings and McElhinny, 1976) were used to select the optimum demagnetization ranges; one specimen from each core (~60% of all specimens) was AF demagnetized, and the remaining specimens were thermally demagnetized. Mean directions and virtual geomagnetic pole (VGP) positions were calculated for all sites. The results from the 12 sites were used to calculate the overall mean paleomagnetic pole for the capping sequence. Table 1 summarizes the paleomagnetic data from the 12 sites of the capping sequence. Only one site (SA 4), which corresponds to the rhyolite flow immediately above the first conglomerate, is of normal polarity; the remaining 11 sites are all reversely magnetized. The reversed and normal directions are tightly grouped, except for the youngest site (SA 12) which is displaced some 25° to 30° (Fig. 3).

DISCUSSION

The mean pole position for the capping sequence is included in Table 2, together with mean pole positions for "stable" North America from other studies. This pole position agrees with those of northern Mexico and seems different from those of stable North America although the angular differences are small (no overlapping of their 95% confidence limits). The total number of results of this study is not great, but the precisions are high, and it is considered that the pole is a reliable estimate for the capping sequence and that the angular discrepancy may be real. The capping sequence pole lies between the Late Cretaceous mean pole (N2, 68.3°N, 160.9°E) and the Oligocene mean pole (N4, 70.5°N, 168.1°E) of the apparent polar-wander path for Mexico (Urrutia-Fucugauchi, 1979b). Although the Mexican apparent polar-wander path is, in general, of similar shape (the post-Jurassic segment) to those proposed for the rest of North America (Irving, 1979; Van Alstine and de Boer, 1978), there are certain consistent differences. These could be due to the preliminary nature of the Mexican path and the general difficulties in constructing such paths. Alternatively, they may indicate tectonic movements between Mexico and stable North America. Results from the Oligocene (Urrutia-Fucugauchi and Pal, 1977) and the Miocene (Urrutia-Fucugauchi and Pal, 1976) have suggested the possibility of a counterclockwise tectonic rotation of northern Mexico relative to the rest of North America. The result from the capping sequence further supports that inference. The amount of movement is, however, near the limit of resolution on the basis of the data available.

The data support a counterclockwise rotation of northern Mexico relative to the rest of North America during early Tertiary time (Fig. 4d). There are several suggested major discontinuities in southern United States–northern Mexico (for example, Albritton and Smith, 1957; Muehlberger and Wiley, 1970; de Cserna, 1976; Silver and Anderson, 1974; Walper, 1980). Although many more studies are required to locate the discontinuity along which the movement took place, at present, the Texas lineament seems a possible zone. It is worth noting that a counterclockwise rotation may restore the continuity of the Cordilleran belt, bringing together the segments of northern Chihuahua and southern Nevada (Fig. 1). An attempt to quantify the tectonic movement by using the Eocene pole

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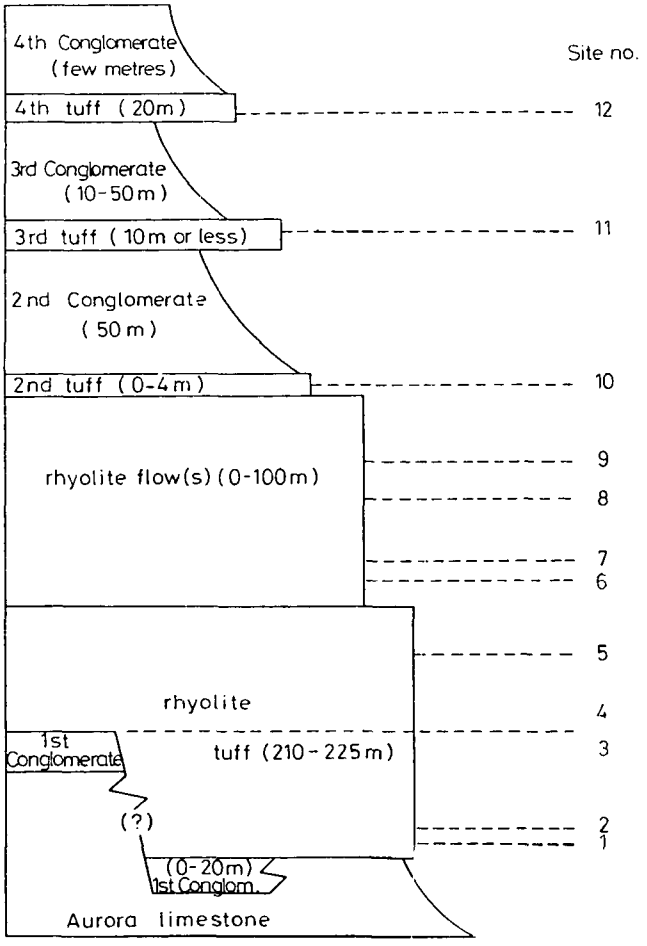


Figure 2. Units forming Capping Series in the Santo Domingo-San Antonio area, Chihuahua, Mexico. These lie on top of Aurora limestone of Cretaceous age. Approximate stratigraphic location of sampling sites is also shown. Figure based on data given by Spurr (1911) and Hewitt (1966).

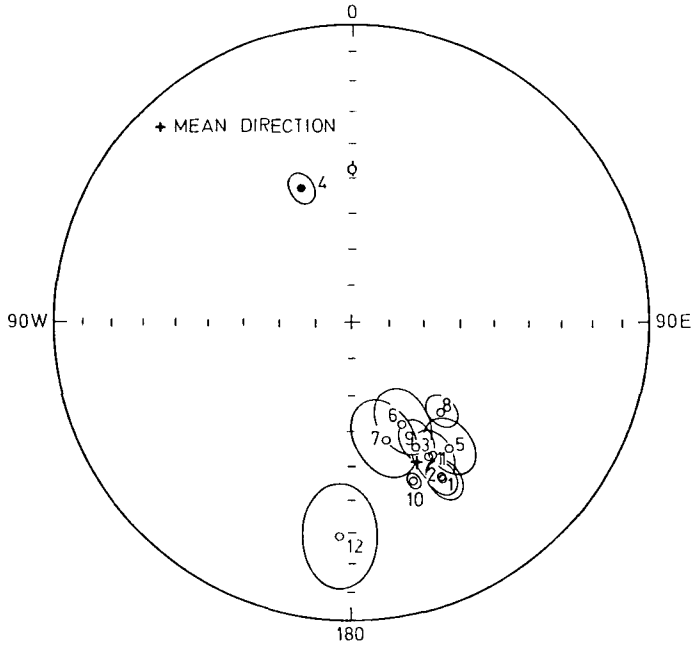


Figure 3. Site mean directions and 95% confidence limits for 12 sites of Capping Series (see Table 1).

TABLE 1. SUMMARY OF PALEOMAGNETIC DATA OF CAPPING SERIES FROM SAN ANTONIO, CHIHUAHUA, MEXICO

Site	Cleaned remanent magnetization N	E	D	I	α_{95}	k	AF (Oe)	T (°C)	Polar Coordinates Lat(°N)	Long(°E)	A_{95}	k
SA 1	3	6	148.5	-39.0	6.6	105	200-350	305	60.9	159.0	5.6	144
SA 2	3	6	148.5	-39.6	5.8	135	200-350	305	60.9	159.8	4.9	187
SA 3	3	5	149.1	-46.5	7.3	112	300	210	62.8	170.1	9.0	73
SA 4	3	5	340.1	50.5	4.0	366	400	210	72.6	178.0	4.8	256
SA 5	3	5	141.4	-45.9	8.7	78	200	305	56.1	171.8	10.0	59
SA 6	2	5	154.4	-59.2	9.3	69	350	210	66.0	201.7	12.4	13
SA 7	2	5	162.8	-56.0	10.7	52	350	210	73.1	201.4	13.3	34
SA 8	3	5	134.8	-55.2	5.1	225	200	305	51.6	186.8	6.5	138
SA 9	2	4	152.3	-51.7	7.2	162	200	305	65.9	178.4	7.3	161
SA 10	4	7	159.1	-43.1	2.2	758	200	305	71.1	158.4	2.1	818
SA 11	4	6	148.5	-46.4	5.6	146	200	305	62.5	170.1	6.4	109
SA 12	4	6	183.1	-28.8	16.5	18	400	305	76.5	58.6	17.4	16
MEAN	12	-	154.1	-47.4	6.5	45	-	-	67.1	172.5	7.0	39
	36	-	154.6	-46.6	4.3	32	-	-	67.4	170.5	4.6	28
	-	65	154.4	-46.8	3.2	32	-	-	67.3	171.1	3.5	27

Note: N = number of samples; E = number of specimens; D = declinations; I = inclination; α_{95} = semiangle of cone of 95% confidence about mean direction (mean pole); k = concentration parameter; AF = alternating field used for bulk demagnetization; T = temperature used for bulk demagnetization.

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position determined here and the mean Oligocene pole from the Mexican apparent polar-wander path (Urrutia-Fucugauchi, 1979a, 1979b) with the mean poles for the late Eocene and Oligocene from Irving's (1979) apparent polar-wander path (item 1 in Table 2) is shown in Figure 4. The apparent polar-wander path of Irving (1979) is the curve that includes the greatest number of poles of all available North American apparent polar-wander paths. According to this matching of the polar-wander path segments (Fig. 4b), the major part of the relative tectonic rotation occurred in post-Oligocene time, and possible earlier movements are pre-Eocene. When alternate estimates of the Eocene and Oligocene pole positions for North America (see Table 2) are used, the tectonic rotation is not determined in longitude, because it is not possible to match the corresponding segments of the apparent polar-wander paths. In these cases, different amounts of tectonic movement occurred during Eocene and Oligocene time, although the major part of the relative rotation remains post-Oligocene. More studies are needed in order to establish the presence and exact timing of the rotation and the nature of any earlier tectonic movements. The main features of the proposed model, presented in a very simplified way in Figure 4, are as follows: The reconstruction shown in Figure 4b was obtained by matching the corresponding Eocene and Oligocene pole positions for Mexico and other parts of North America and restoring the continuity of the Cordilleran orogenic belt. The agreement of the pole positions is excellent, which supports the validity of the reconstruction. At about the end of the Eocene some 40 m.y. ago, the absolute rate of motion of the North American plate decreased (Coney, 1972, 1978). At that time the direction of the Pacific plate changed from predominantly northward (between 80 and 40 m.y. B.P.) to northwestward (after 40 m.y. B.P.; Jarrard and Clague, 1977; Coney, 1978). Also, during the early Tertiary, major events were affecting the southwestern margin of North America, with the subduction of the Farallon ridge, transformation of the plate boundaries, and general plate reorganization (Atwater, 1970; Coney, 1978). Figures 4a and 4c are extrapolations based on Figures 4b and 4d. The proposed tectonic rotation may also explain the dislocation of the Mesozoic batholith of western Mexico (Figs. 1, 4), which once may have extended from southwestern Jalisco into the Peninsular Ranges of southern California (Gastil and others, 1975; Gastil and Jensky, 1973). The model also provides a possible explanation for the formation of the extensive deformed region and the orogenic belt bend occurring from south of Monterrey to western Coahuila, where the trend of the orogenic belt (locally the Sierra Madre Oriental) changes from predominantly north-northwest to west (Fig. 1). From western Coahuila the belt trend changes again from west to northwest and continues into Chihuahua (de Cserna, 1956; Tardy, 1975; King, 1969). The structures may have formed by deformation against the Coahuila peninsula. Tardy (1975) indicated that the deformation and generation of the great recumbent folds, which formed great nappe structures, occurred during the Laramide orogeny (early Tertiary) and that the deformation direction in the Parras nappe is north-northeast, which agrees with the proposed model (Fig. 4).

Within Mexico (Fig. 1) the surficial expression of the Cordilleran orogenic belt is obscured by the younger and extensive evidence of volcanism, the Mexican volcanic belt, which perhaps marks yet another major discontinuity of the orogenic belt (Dengo, 1975). Some paleomagnetic evidence (Urrutia-Fucugauchi, 1981) from an area immediately to the south of the Mexican volcanic belt suggests the occurrence of relative motions between northern and southern Mexico, north and south of the Mexican volcanic belt. South of this belt, the regional trend and characteristics of the orogenic belt appear to change (Fig. 1). It is possible to speculate that tectonic movements also took place along the trace of the Mexican volcanic belt during Late Cretaceous and early Tertiary time. Gastil and Jensky (1973) suggested a 175-km movement

TABLE 2. SUMMARY OF POLE POSITIONS FOR THE EARLY TERTIARY OF NORTH AMERICA

Pole explanation	No. of sites	Pole Lat (°N)	Pole Long (°E)	Pole A_{95}	Reference
1. Pole positions for stable North America					Irving (1979)
(30 m.y.) Oligocene	18	86	166	5	
(40 m.y.) Late Eocene	16	83	164	5	
2. Pole positions for intervals 2 and 3 (mode poles)					Van Alstine and de Boer (1978)
Late Eocene-early Miocene	6	86	137	5	
Late early Paleocene-Middle Eocene	7	87	220	9	
3. Pole position for North American upper Tertiary rocks	3	85	114	6	Beck (1976)
4. Pole position for Eocene North American rocks	4	87	169	6	Simpson and Cox (1977)
Expected North American Eocene pole from data of North America, Europe and USSR	19	78	187	5	
5. Pole position for Oligocene of Mexico	4	71	168	11	Urrutia-Fucugauchi (1979a)
6. Capping Sequence Chihuahua, northern Mexico	12	67	171	5	This study

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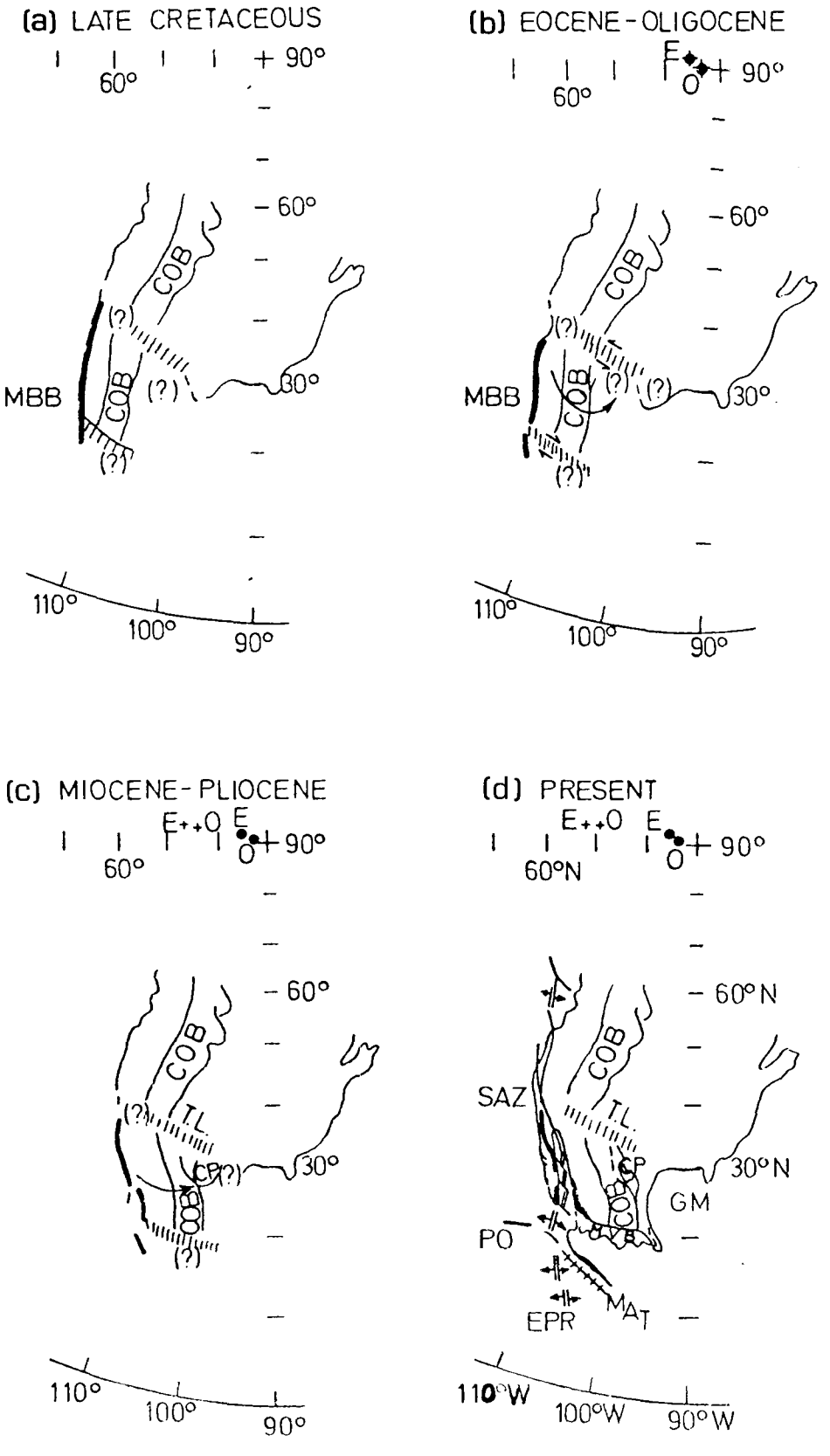


Figure 4. Schematic presentation of tectonic model proposed for evolution of southern North America. Symbols: paleomagnetic pole positions for the Eocene (E) and Oligocene (O) of northern Mexico (+) and North America (•); COB, Cordilleran orogenic belt; TL, Texas lineament; CP, Coahuila peninsula; MVB, Mexican volcanic belt; MBB, Mesozoic batholith belt of western Mexico; MAT, Middle America Trench; PO, Pacific Ocean; SAZ, San Andreas transform fault zone. Present-day coordinates and coastal lines of (d) kept in (a), (b), and (c) for sake of presentation. North America is kept fixed for relative tectonic motion. See Figure 1, Table 2, and text.

along the trace of the volcanic belt during the Late Cretaceous and early Tertiary and a further 260-km displacement during the Miocene or Pliocene, both in the direction and sense required by the model proposed here.

A tectonic rotation of northern Mexico seems supported by the paleomagnetic results. The tectonic model proposed is still highly speculative, on the basis of the available data; however, it can be tested by obtaining further paleomagnetic data from other areas, of both similar and different ages, to determine whether the apparent pattern is real or fortuitous. Additionally, the model provides a basis for comparing geological and geophysical information. Gose and others (1979) have found paleomagnetic evidence for a 125° to 130° counterclockwise rotation of northern Mexico between the Triassic and Early Jurassic and possibly continuing into the Cretaceous. This may have taken place along a zone approximately between the United States and Mexico, possibly coinciding with the Texas lineament (Gose and others, 1980).

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